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A Review on AI Driven Distributed SDN for Energy -Efficient Routing in Large Scale IoT Network

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Abstract

The IIoT networks are long-standing problems that include high energy requirements, network clogging, and inefficiency on scalability, especially when centralized SDN is used. These challenges slowdown the functionality of IIoT systems to deliver feasible and real-time performance in large-scale applications. In order to overcome these limitations, the current paper provides a systematic literature review of 52 studies devoted to the topic of integrating Artificial Intelligence (AI) with distributed SDN models. It is found that dynamic routing, adaptive cluster-head selection, and mobile sink path planning are possible through AI techniques, including Reinforcement Learning (RL), Genetic Algorithms (GA), swarm-based optimization and fuzzy logic. Simulation-based analyses have uniformly demonstrated large performance-improvement, such as up to 78 % reductions in energy usage, 63 % increases in network lifetime, and significant reductions in latency and packet loss over traditional centralized SDN systems. The review also presents new research directions, including AI-aided load balancing, real-time traffic optimisation and hybrid AI-SDN systems with the integration of blockchain and edge computing. In general, the results prove that AI-driven distributed SDN frameworks are a scalable and energy-efficient way to solve the bottlenecks of centralized architectures, which is a feasible solution to resilient and adaptive IIoT networks in smart cities, manufacturing, and healthcare.

Keywords: AI-driven SDN, Industrial Internet of Things (IIoT), Energy-efficient routing, Distributed SDN, Optimal Data Gathering

I. INTRODUCTION

Industrial Internet of Things (IIoT) is a paradigm shift that is revolutionizing the modern industries by combining massive networks of devices, sensors, and systems that interact and share real-time data. This connectivity leads to better decision-making, automation, and operational efficiency, and, therefore, makes IIoT a necessity in various fields such as smart cities, manufacturing, and healthcare. In smart-city systems, the IIoT can enable management of key infrastructure, such as traffic systems, energy grids, and waste management, among others, and thus create more sustainable and efficient urban environments.

In manufacturing, IIoT enhances production by ensuring predictive maintenance, automation of processes, and quality management, therefore, adding productivity to manufacturing and reducing expenses. The healthcare industry relies on IIoT as the foundation of remote patient monitoring, wearable sensor implementation, and advanced medical data analytics, all of which culminate in higher patient outcomes and an improved flow of clinical work [1-5]. This is further aggravated by the fact that battery-powered components have limited energy supply and therefore energy-efficient routing algorithms need to be designed to balance the load in the network [6, 7]. Similar issues have been raised by researchers in terms of reliability of communication across a sensor-rich, low-power environment, which is why adaptive load-balancing approaches are necessary [8, 9]. The growth of the Internet-of-Things (IoT) ecosystem is characterized by the increased data traffic, which, in its turn, triggers the occurrence of network congestion, latency, packet loss, and reduced overall reliability. With the expansion of these networks, maintaining performance in the presence of a multiplication of connected devices remains a constant challenge [10]. It is essential to overcome this challenge to achieve the potential of the IIoT systems in modern industrial settings.

A. Role of Software-Defined Networking (SDN) in IIoT

The potential use of the Industrial Internet of Things (IIoT) in areas like smart cities, manufacturing and healthcare is significant; however, the intricacies of the deployment and operation of large-scale, resource-constrained networks are still significant. The most prominent of these obstacles are limitations related to energy efficiency, network congestion and scalability, which become more pronounced when the density of connected devices rises. To solve such complexities, Software-Defined Networking (SDN) has been proposed as a potential solution, which offers centralized control and dynamic reconfiguration of the network to coordinate and optimize the performance of such environments [11-13]. However, the weaknesses of centralized SDN, namely, the high latency, limited scalability, and single-point-of-failure vulnerability, still make the process of managing IIoT infrastructures a challenge.The study conducted by Siddiqui et al. (2022) [14] conducted in-depth research to determine the level at which the Software-Defined Networking (SDN) may enhance Industrial Internet of Things (IIoT) systems. Through a methodical overview of over 300 research articles, the authors show that SDN can significantly increase network scalability, provide centralized control, and offer network programmability all of which are impossible to go without in the dynamic complex IIoT environment. However, the paper also points out that significant practical barriers, such as real-time responsiveness, interoperability, and security, still exist, and they prevent the full adoption of SDN-based IIoT deployments.The major drawbacks of Software-Defined Networking (SDN) do not allow it to support real-time variation fully and respond to the increasing demands of Industrial Internet of Things (IIoT) environments, which drives the exploration of distributed SDN architectures and the integration of Artificial Intelligence (AI) to support

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dynamic adaptation and lead to greater energy efficiency in IIoT systems [15-17].

B. *The Need for AI Integration with SDN*

The inefficiency of centralized Software-Defined Networks (SDN) to manage the complexity and dynamic nature of large-scale Industrial Internet of Things (IIoT) systems highlights the need of intelligent and flexible network management systems. With the proliferation of IIoT networks, the traditional SDN setups start to fail in delivering the fast, energy-efficient routing, load balancing, and scaling adaptation required to achieve optimal operational efficiency. AI based techniques, especially Reinforcement Learning (RL) and Genetic Algorithms (GA), thus provide better mechanisms of optimizing the network performance by deriving insights out of real-time network states and dynamically adjusting routing behaviors [18, 19]. Artificial intelligence and software-defined networking are converging to facilitate adaptive response to node failure, energy depletion and traffic congestion and increase the resilience and efficiency of network operation. Moreover, some significant efficiencies can be observed when AI is used in combination with distributed SDN systems compared to traditional centralized architectures. Specifically, distributed SDN involves the use of several SDN controllers that collaborate in controlling individual portions of the network, thus providing better scalability and greater fault tolerance [20,21]. Alhilali et al. (2023) [22] conducted a systematic survey with the purpose of investigating the appropriateness of Artificial Intelligence (AI) in enhancing load balancing in Software-Defined Networking (SDN) systems. They investigated three major AI paradigms, such as machine learning, deep learning and reinforcement learning, and their use in traffic management and optimization of resources. The findings were that traditional SDN faces significant limitations in supporting dynamic network environments and providing real time decisions. The study thus concluded that the integration of AI enables intelligent, scalable, and autonomous load balancing, which makes AI an essential component of future-proof network infrastructures. The combination of distributed SDN and AI algorithms includes real-time network optimization and enables smart decision-making on local and global levels.

The current literature review examines how Artificial Intelligence (AI) and Software-Defined Networking (SDN) are becoming integrated in Industrial Internet of Things (IIoT) systems. As the number of IIoT-connected devices increases and the network becomes more complex, it becomes more challenging to coordinate an increasingly large number of network-connected devices and maintain network performance. The traditional centralized SDN designs often face the challenge of managing the dynamic nature of these environments, which has led to the emergence of issues like energy inefficiency, network congestion and lack of scalability. In comparison, AI-based methods, especially Reinforcement Learning (RL), Genetic Algorithm (GA), and other pertinent methods demonstrate great potential in optimizing routing, load balancing, and energy management. The current review examines the AI-driven SDN platforms that were developed to overcome the drawbacks of IIoT networks. The combined use of continuous real-time monitoring of the network conditions with data-driven decision-making, allows AI-driven SDN to enhance energy consumption optimization, latency minimization and scalability, which are the features that are of utmost importance when using IIoT in smart-city settings, industrial automation, and healthcare. Further, the paper evaluates the effectiveness of AI-based SDN architectures in energy efficient routing where battery powered IoT devices and sensor networks can work long durations without the degradation of their performance to a large extent. The limitations of the existing AI-SDN models will be also a subject of the review, along with research gaps and suggestions to developments required to fully utilize the power of AI in the management of large-scale IIoT systems. Finally, the purpose of the review is to provide a detailed perception of the way artificial intelligence-based SDN could redefine IIoT networks with its ability to provide network resilience, energy efficiency, and real-time adaptability.

II. BIBLIOMETRIC SURVEY

To systematically assess the scope, trends, and thematic directions of current research in Software Defined Networking (SDN) applied to the Industrial Internet of Things (IIoT) and Smart Cities, a bibliometric analysis was conducted using the Scopus database. The search query used was:

("Software Defined Network*" OR "SDN*" AND ("Industrial IoT" OR "IIoT" OR "Smart Cities")

Such search approach provided a rich data set on which the current survey is based. Along with the thematic and geographic analysis, the screening procedure of documents based on the PRISMA was also carried out to narrow down the articles included in the qualitative synthesis in the further literature review. To improve transparency and methodological rigor during the selection of articles, the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) protocol was chosen [23-25]. The initial number of retrieved records was 906 as shown in Figure 1. Upon elimination of duplicates, 873 unique articles were left. They were selected by title and abstract, and 210 records were selected as potentially relevant. After the full-text assessment, 90 articles were evaluated as eligible, and 52 were used in the final thematic review.

The current systematic filtering process ensures that the literature covered in the sections below is consistent with thematic relevance as well as methodological quality.From the original 906 retrieved documents, year-wise analysis reveals a significant growth trajectory.

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Figure 1. PRISMA flow diagram showing document screening and selection process

As shown in Figure 2, research activity remained sparse until 2016, after which publication frequency increased sharply peaking at 118 publications in both 2022 and 2024. The current year (2025) shows 40 publications at the time of analysis, suggesting continued interest. This growth pattern mirrors the increasing demand for programmable, scalable, and energy-aware networking solutions, particularly in the context of complex industrial and urban systems.

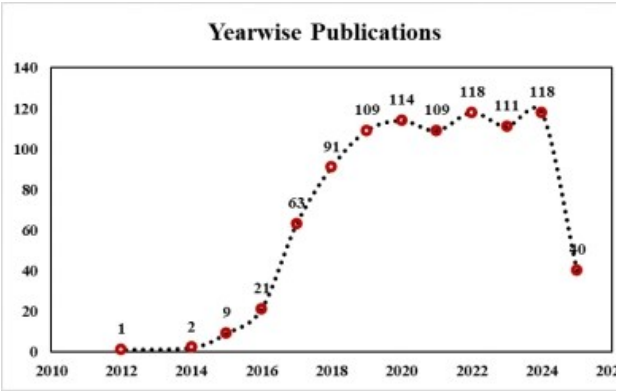

Figure 2. Year-wise distribution of publications related to SDN in IIoT and Smart Cities (2012–2025)

To identify dominant research themes, a keyword co-occurrence network was generated using VOSviewer [26,27]. Figure 3 reveals several well-structured clusters; Red Cluster (AI and Learning Systems) covers *Artificial Intelligence*, *Learning Systems*, *Active Learning*—highlighting integration of adaptive algorithms in SDN environments. Green Cluster (Optimization Methods) includes *Adaptive Boosting*, *Feature Selection*, and *Discriminant Analysis*, emphasizing efficiency-driven approaches. Yellow Cluster (Network Architectures) centers on *Cluster-Heads*, *Internet Protocols*, and *Cluster Computing*, relating to SDN-enabled hierarchical routing. Blue Cluster (Industrial Deployment) terms such as *Industrial Internet of Things* and *Computation Offloading* show focus on SDN's applicability in IIoT. Orange Cluster (Security and Blockchain) addresses emerging interest in trust, cybersecurity, and decentralization.

This distribution affirms the field's interdisciplinary orientation, spanning AI, optimization, security, and industrial automation.

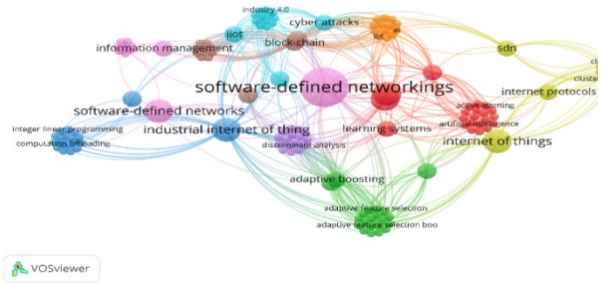

Figure 3. Keyword co-occurrence network visualized using VOSviewer, illustrating thematic clusters in SDN-related IIoT and Smart City research.

An analysis of author affiliations indicates a broad global research footprint. As presented in Figure 4, China (183 publications), India (168), and the United States (99) emerged as the top contributors. Substantial contributions also came from countries such as the United Kingdom, Canada, Pakistan, South Korea, and Saudi Arabia. The presence of both developed and emerging economies in this domain illustrates the global recognition of SDN's potential in transforming industrial networks and smart infrastructure. The geographic spread also indicates regional investments in future-ready networking technologies.

Figure 4. Global distribution of research publications on SDN in IIoT and Smart Cities (2012–2025)

The bibliometric results highlight a mature and expanding body of literature with clear thematic clustering around AI-enhanced SDN, cluster-based optimization, and secure, scalable architectures. The global and temporal trends suggest a collective movement toward real-time, energy-efficient, and adaptive communication systems—reinforcing the need for a deeper examination of how these technologies converge. These insights directly inform the structure of the forthcoming Literature Review (Section 3), which delves into the specific innovations, limitations,

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and research gaps in centralized SDN, AI-based clustering, mobile sink path planning, and AI-integrated distributed frameworks.

Based on the literature review, it is clear that even though AI-enhanced SDN can lead to better routing efficiency, better energy control, and dynamic adaptability in IIoT, there are still some gaps. The literature published is mainly simulation-based with little validation in the real-world resource-limited settings. The combination of SDN cluster-head selection, mobile sink path planning, and distributed SDN controllers have been discussed separately, whereas in practice they are highly interdependent. What is more, the computational and energy cost of AI models at the edge is seldom measured, and this raises doubts about their viability in IIoT devices. Lastly, the majority of the methods are single-objective-oriented (e.g., energy minimization), as trade-offs between latency, reliability, fairness, and scalability are not systematically explored. The gaps point to the necessity of AI-based distributed SDN frameworks that mutually trade-off energy efficiency, latency, and scalability with realistic IIoT constraints, with the assistance of extensive benchmarking and reproducible evaluation protocols.

III. AI-Driven SDN Framework for IIoT Networks

A. Limitations of Traditional Centralized SDN in IIoT

While Software-Defined Networking (SDN) offers centralized control and flexibility, it faces significant limitations in managing large-scale Industrial IoT (IIoT) networks.The primary issue is scalability, as the central controller becomes a bottleneck when the number of devices grows. This leads to increased latency and delayed responses to network changes, affecting network performance. Another limitation is energy inefficiency, as centralized SDN struggles to optimize energy usage due to reliance on a single control point for decision-making, often leading to suboptimal energy consumption. Additionally, latency becomes a critical problem in real-time applications, where centralized SDN introduces delays due to all decisions routing through the central controller. Lastly, fault tolerance is compromised, as the central controller represents a single point of failure. These issues necessitate exploring distributed SDN combined with AI to enhance scalability, adaptability, and energy efficiency.

Udayaprasad et al. (2024) [28] aimed to overcome the scalability and energy limitations of traditional SDN in large-scale IIoT networks. They used simulation-based performance analysis with AI algorithms such as GA, PSO, and ABC integrated into a distributed SDN model. Their methodology focused on optimizing routing and load balancing in a decentralized architecture. Major findings included a 63% improvement in network lifetime and a 78% reduction in energy consumption. The study concluded that AI-driven distributed SDN significantly outperforms centralized SDN models in terms of adaptability and energy efficiency. Tcherak et al. (2023) [29] investigated energy-efficient routing in SDN-based wireless sensor networks. They employed a simulation in the SDN-WISE environment and used the Naive Bayes algorithm for probabilistic traffic routing. The goal was to overcome the rigidity of static routing in centralized SDN. Their findings showed that the model improved packet delivery and network lifetime over traditional Dijkstra routing. The study concluded that intelligent, learning-based routing strategies can significantly enhance the energy performance of SDN in IIoT environments compared to static approaches.

Nazari et al. (2023) [30] focused on developing an energy-aware routing algorithm—fuzzy-IAVOA—for SDN-enabled IoT environments. Their simulated study combined fuzzy logic with optimization strategies to dynamically cluster nodes and manage traffic. The method addressed the inflexibility of traditional SDN in adapting to energy levels and traffic loads. Their findings showed improved QoS, reduced latency, and better energy utilization. The conclusion emphasized the benefit of AI-based techniques in overcoming the static nature of centralized SDN by introducing context-aware, energy-sensitive routing. Kamboj and Pal (2022) [31] explored energy-aware routing in SDN-enabled data centers using Integer Linear Programming (ILP) and heuristic methods. Simulating on Fat-Tree and BCube topologies, they dynamically adjusted routing paths based on traffic demands. Their results showed energy savings of 17–24% along with improved QoS. Though focused on data centers, the findings underscore the inefficiency of static routing rules in centralized SDN and highlight how heuristic optimizations can partially mitigate energy and performance limitations in large-scale deployments. Walubita et al. (2023) [32] explored energy efficiency through intelligent rule placement in SDN controllers. They proposed a heuristic-based model to optimize flow rule insertion, which they validated through simulation. The study found that reducing the number of installed flow rules minimized power consumption and improved controller performance. Their conclusion stressed that centralized SDN systems must evolve to include rule optimization to support energy-efficient and scalable IIoT deployments.

B. AI-Based Cluster Head Selection for Energy Efficiency

In Industrial IoT (IIoT) networks, the efficient selection of Cluster Heads (CHs) plays a pivotal role in optimizing energy consumption and extending the network lifetime. Cluster Heads are responsible for aggregating data from the sensor nodes and forwarding it to the sink or central network, thereby reducing overall communication overhead and energy usage. However, traditional methods of Cluster Head selection often rely on static or predefined approaches, which fail to adapt to the dynamic nature of IIoT environments. These methods can lead to imbalanced energy consumption, where certain Cluster Heads deplete their batteries faster than others, resulting in early node failure and compromised network performance. In Industrial IoT (IIoT) networks, Cluster Head (CH) selection is crucial for energy efficiency and communication optimization. Traditional static methods lead to imbalanced energy consumption, with some nodes depleting faster than others. AI-based algorithms, such as Genetic Algorithms (GA) and Reinforcement Learning (RL), provide dynamic solutions by selecting Cluster Heads based on real-time energy levels, node proximity, and load balancing. These AI techniques improve energy efficiency, ensure even traffic distribution, and enhance network longevity by reducing overload and improving scalability, outperforming traditional static methods.

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Kuila and Jana (2014) [33] focused on developing an energy-efficient clustering algorithm for wireless sensor networks using a GA-based approach. Through simulations, the algorithm selected CHs based on node energy and load metrics. Their findings demonstrated a significant enhancement in network lifetime and balanced energy consumption across nodes. The study concluded that GA is a powerful optimization tool for improving both energy distribution and network stability in WSNs Hu et al. (2009) [34] aimed to improve energy efficiency in Wireless Sensor Networks (WSNs) through a fuzzy logic-based cluster head (CH) selection mechanism. The methodology involved selecting CHs based on remaining energy and node density, contrasting the LEACH protocol's random model. Simulations demonstrated that the proposed NECHS protocol significantly outperformed LEACH, with improvements ranging from 0.09% to 26.8% across various network death metrics. The study concludes that fuzzy logic enhances decision-making in CH selection, leading to improved energy efficiency and longer network lifetimes. Thein and Thein (2010) [35] developed an adaptive CH selection algorithm based on residual energy to balance energy load in WSNs. The study enhanced LEACH's stochastic method by modifying node selection probabilities. Simulations confirmed that their model improved energy distribution and prolonged network lifespan more effectively than traditional approaches. Their methodology emphasized dynamic cluster adaptation and regular CH rotation. The conclusion asserts that intelligent energy-aware selection significantly improves WSN performance and longevity.

Reddy et al. (2023) [36] proposed an enhanced Grey Wolf Optimization algorithm (EECHIGWO) to optimize CH selection in IIoT-based WSNs. Using parameters like sink distance and residual energy, the method aimed to maximize network lifetime, stability, and throughput. The study addressed common limitations in GWO such as poor convergence and lack of diversity. Simulation results showed that EECHIGWO achieved 169%–333% improvement over other protocols in stability and energy efficiency. The conclusion supports the effectiveness of swarm intelligence for intelligent CH selection in dynamic WSN environments. Kaur et al. (2018) [37] introduced a Flower Pollination Algorithm (FPA)-based method for CH selection to enhance energy conservation in WSNs. The simulation-based evaluation compared this approach with LEACH, C-LEACH, and K-Means protocols across metrics such as node longevity, energy dissipation, and packet loss. Results showed that FPA led to significant improvements in energy efficiency and network stability. The conclusion underlined the potential of biologically inspired algorithms for adaptive and efficient cluster formation in sensor networks. Chaturvedi et al. (2016) [38] aimed to optimize CH selection through a cross-layer energy-based algorithm over the LEACH protocol. The methodology integrated node energy metrics and cross-layer data to improve the reliability and lifetime of WSNs. Simulations confirmed that the proposed model enhanced energy efficiency and stability by selecting CHs based on real-time energy availability. The study concluded that cross-layer optimization is crucial for sustainable WSN deployment, especially in resource-constrained environments.

C.AI-Based Mobile Sink (MS) Path Planning

In Industrial IoT (IIoT) networks, Cluster Heads (CHs) play a critical role in reducing communication overhead by aggregating data from sensor nodes and relaying it to the central system. Traditional methods of Cluster Head selection are often static and inefficient, leading to imbalanced energy consumption, where certain Cluster Heads deplete their energy faster than others. This inefficiency results in early node failures and uneven network load distribution, reducing overall network longevity. To address these challenges, AI-based algorithms such as Genetic Algorithms (GA) and Reinforcement Learning (RL) provide dynamic, energy-efficient solutions. These algorithms evaluate real-time network conditions, including energy levels, node proximity, and traffic distribution, to intelligently select Cluster Heads that optimize energy consumption and ensure a balanced network load. The AI-based approach outperforms traditional methods by selecting Cluster Heads based on the current energy state and local network conditions, leading to more efficient routing and a longer network lifetime. Compared to static methods, AI-driven cluster head selection enhances network scalability and stability, allowing for real-time adaptation to changing network conditions. Future research could explore real-time adaptive clustering using Reinforcement Learning, which would further enhance energy efficiency and network performance in dynamic IIoT environments.

Al-Janabi et al. (2018) [39] aimed to improve mobile sink scheduling in large-scale IIoT using AI techniques. Their simulation used GA and PSO to optimize sink movement based on energy and delay factors. The mobile sink path was dynamically planned to reduce latency and enhance energy efficiency. Findings indicated a 54% improvement in network lifetime and 61% reduction in delay. The study concluded that integrating AI for sink path optimization enhances performance in IIoT networks and overcomes limitations of static sink models. Wu and Liang (2019) [40] aimed to address the challenges of path planning for mobile sinks in Mobile Wireless Sensor Networks (MWSNs). Through a comprehensive survey of optimization schemes, the study compared existing strategies including heuristic and AI-based techniques. The methodology involved classifying and analyzing different path planning methods and identifying their limitations in energy efficiency and coverage. The study found that existing algorithms often lack scalability and real-time responsiveness. It concluded that AI-based multi-objective approaches could offer promising future directions for dynamic and efficient path planning. Liu et al. (2022) [41] focused on secure and accurate data aggregation in WSN-based IIoT using a single mobile sink. They proposed a trust-based model (WIM) that integrated robust aggregation trees and outlier elimination within a virtual grid. Their methodology included real-time simulations with mobile sink mobility patterns. Major findings demonstrated enhanced data accuracy and reduced delay, while maintaining network robustness under random sink movements. The study concluded that incorporating trust mechanisms with AI-based mobility scheduling significantly improves data integrity and network performance.

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Mishra et al. (2016) [42] proposed a delay-bound rendezvous-point-based path planning method for mobile sinks in WSNs. Their simulation-based methodology used hexagonal partitioning of the network and optimized rendezvous point selection using network parameters. The study aimed to reduce path length while respecting delay constraints. Results showed that their algorithm outperformed existing models in terms of hop count, lifetime, and delay. The study concluded that AI-augmented delay-aware planning enhances sink mobility efficiency and mitigates hotspot issues. Chang et al. (2020) [43] developed a novel AI-based mobile sink path planning scheme for large-scale WSNs. The methodology used angle bisector-based algorithms and inner center routing with obstacle awareness. It aimed to reduce sink travel distance and improve network lifespan. Major findings included a 1.1 km reduction in average moving distance compared to heuristic approaches, especially in dense sensor deployments. The study concluded that adaptive and AI-driven path planning significantly improves efficiency and is scalable for real-world applications. Donta et al. (2020) [44] proposed an extended Ant Colony Optimization (ACO) method for planning mobile sink paths in event-driven WSNs. The methodology involved selecting optimal rendezvous points (RPs) and integrating virtual RPs to balance energy loads. Simulation results demonstrated reduced data loss and prolonged network lifetime. Findings emphasized the importance of adaptive RP selection and sink mobility strategies for efficient data aggregation. The study concluded that AI-based techniques like ACO enhance responsiveness and performance in dynamic WSN environments. Xie et al. (2024) [45] aimed to improve mobile sink path optimization in wireless sensor networks by addressing power consumption and route efficiency. The authors proposed the ACO-TU algorithm, an extension of the Ant Colony Optimization (ACO) model, which integrates node pheromone dynamics and grouping strategies for mobile units. Using a simulation-based methodology, the study demonstrated that the algorithm significantly reduces both the distance traveled and energy consumption of the mobile sink. The experimental results confirmed the superior optimization performance of the ACO-TU method, validating its effectiveness in extending network lifetime and enhancing the reliability of signal transmission.

D.AI-SDN Integration for Adaptive IIoT

Software-Defined Networking (SDN) offers centralized control and flexibility in managing IIoT networks, allowing for dynamic reconfiguration and efficient resource allocation. However, traditional centralized SDN faces challenges in real-time adaptation due to its inability to respond quickly to dynamic changes in network conditions, such as node failures, traffic congestion, or energy depletion. Integrating Artificial Intelligence (AI) with SDN enhances its capabilities by enabling real-time decision-making and adaptive routing based on network feedback. AI algorithms, such as Reinforcement Learning (RL) and Deep Q-Learning (DQN), allow SDN controllers to continuously learn from the network's behavior and make intelligent routing decisions. This integration enables networks to adapt dynamically to fluctuations in energy levels, traffic patterns, and node availability, ensuring optimal performance even under fluctuating conditions.

AI-based SDN systems improve energy efficiency by optimizing routing paths to minimize energy consumption and avoid congestion. Additionally, they provide load balancing by distributing traffic evenly across the network, preventing node overload and ensuring efficient utilization of resources. Compared to traditional SDN, the AI-SDN framework offers a more resilient and scalable solution for managing large-scale IIoT networks, where real-time adaptability is crucial to ensuring consistent and reliable network operation.

Jiang et al. (2023) [46] aimed to explore how AI-enabled SDN architectures enhance security and scalability in IIoT. The study reviewed state-of-the-art research and proposed an architecture dividing IIoT networks into three layers, integrating technologies like blockchain, edge AI, and time-sensitive networking. By analyzing AI's role in control, automation, and data sharing, the study found AI-SDN integration significantly improves flexibility, real-time responsiveness, and resilience in IIoT environments. This convergence was found to be essential for future secure, autonomous industrial systems. Fernández-Fernández et al. (2016) [47] aimed to enhance energy efficiency in SDN using AI-based traffic engineering. Their simulation used real topologies and optimization models to evaluate performance. The AI-enabled SDN controller minimized active links based on traffic constraints, which led to significant energy savings. The findings demonstrated a 60% energy reduction while maintaining high network performance. The study concluded that incorporating heuristic traffic-aware AI control can significantly optimize energy usage in SDN environments without compromising service quality. Li et al. (2018) [48] proposed an adaptive SDN-based IIoT transmission architecture incorporating edge computing. The methodology classified data streams by urgency and applied coarse- or fine-grained routing strategies using path difference degree (PDD) and power adaptation. Simulation results showed significant improvements in latency, throughput, and energy efficiency. The hybrid architecture effectively supported real-time constraints and dynamic network conditions, making it suitable for industrial environments requiring differentiated QoS. Josbert et al. (2024) [49] conducted a comprehensive survey on SDN integration in smart IIoT systems. They proposed a hierarchical control-plane architecture incorporating AI, DHT, AdaptFlow, and edge/cloud storages. The study highlighted flow installation, traffic routing, and fault tolerance as critical challenges. AI/ML-based techniques were found to improve performance across these domains. The proposed model demonstrated promise in real-time response, scalability, and energy efficiency in IIoT settings.

Maddu & Rao (2025) [50] introduced a lightweight and adaptive AI framework for intrusion detection in SDN-based IIoT. Using EfficientNet-B0, autoencoders, and DQN for real-time mitigation, the model achieved 96% detection accuracy and <15 ms latency. The methodology combined supervised and self-supervised learning, elastic weight consolidation, and Tiny-YOLO for edge deployment. The integrated system ensured robust, low-latency, and scalable protection for large-scale IIoT networks. Dehkordi (2025) [51] proposed the EDBLSD-IIoT hybrid architecture

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integrating AI, SDN, blockchain, and edge computing for optimized IIoT performance. Using the Whale Optimization Algorithm (WOA) for cluster head selection and traffic management, the framework reduced latency to 0.45 ns and energy usage by 72%. Simulation in a smart car factory validated improvements in throughput and CPU load, outperforming GA and ACO-based models. Ahmed et al. (2024) [52] aimed to compare routing algorithms in SDN using AI to support future networks handling massive IoT data. The study proposed an architecture that integrates AI and blockchain within SDN controllers to ensure secure, efficient routing and traffic management. Through simulation-based evaluation, the authors examined how AI aids the controller in achieving real-time decisions, reducing latency, and improving system performance. The results showed that AI-enabled routing enhances security and responsiveness in complex IoT environments. The paper concluded that AI integration is essential for scalable and intelligent SDN systems.

III. Discussion

The integration of AI with SDN in IIoT networks has gained significant attention, but there are several research gaps that remain unaddressed. While AI has been explored for optimizing energy-efficient routing and real-time adaptation, much of the research focuses on general IoT networks, leaving IIoT-specific challenges such as high node density, real-time data aggregation, and energy management largely unexplored. Additionally, the integration of AI-driven cluster head selection and mobile sink path planning within AI-SDN frameworks remains under-researched, especially when considering the interdependencies between these elements in IIoT environments. Another significant gap lies in the evaluation of performance metrics. While some studies address energy consumption, there is a lack of multi-metric optimization, which considers energy, latency, and load balancing simultaneously. Furthermore, the real-time synchronization of distributed AI-SDN systems has not been thoroughly studied, making it difficult to apply AI-SDN to large, complex IIoT networks in a real-time adaptive manner. Lastly, simulation platforms that can replicate large-scale IIoT environments for testing AI-SDN frameworks are scarce, limiting the ability to validate and refine these approaches in realistic settings. Addressing these gaps is essential for the successful deployment of AI-based SDN solutions in IIoT systems.

A summary of representative studies and their methodologies, performance metrics, and results is provided in Table 1. The table shows that although AI-driven distributed SDN can be shown to consistently enhance energy efficiency and flexibility, most comparisons are simulation-based and do not often take into account multi-objective trade-offs and practical limits.

Table 1 Comparison of existing AI-SDN approaches in IIoT

Study (Year)	Focus / Technique	Key Findings
Udayaprasad et al. (2024) [28]	Distributed SDN with GA, PSO, ABC	78% energy savings; 63% longer lifetime
Tcherak et al. (2023) [29]	SDN-WSN with Naïve Bayes routing	Higher packet delivery; longer network life
Nazari et al. (2023) [30]	Fuzzy-IAVOA for energy-aware routing	Reduced latency; improved QoS and energy balance
Reddy et al. (2023) [36]	Cluster-head selection with Grey Wolf Optimization	169–333% improvement in stability, throughput
Al-Janabi et al. (2018) [39]	Mobile sink scheduling with GA, PSO	54% increase in lifetime; 61% delay reduction
Chang et al. (2020) [43]	AI-based path planning for mobile sink	Shorter travel distance; improved lifetime
Jiang et al. (2023) [46]	AI-SDN with Edge AI and blockchain	Greater resilience; real-time adaptability
Dehkordi (2025) [51]	Hybrid AI-SDN with Whale Optimization	Latency cut to 0.45 ns; 72% energy reduction

IV. LIMITATIONS OF EXISTING REVIEW

Although some encouraging findings have been obtained, the current literature on the use of AI-driven SDN in IIoT exhibits some limitations. The majority of works follow simulation only validation that uses simplified traffic and topology models and thus it is hard to generalize the results to real industrial environment. Cluster-head selection strategies tend to overlook reconfiguration overheads, control signaling costs and scalability in the face of mobility or node churn. The studies of mobile sink path planning normally presuppose favorable circumstances, disregarding the interference and queue behaviour, as well as traffic sensitivity to deadlines. At SDN layer, there is little work on controller synchronization lag, flow-rule churn, or computational complexity of AI models deployed at the edge. Furthermore, most studies are single-objective (primarily energy), whereas the multi-objective trade-off between energy, latency, reliability, and fairness have been less studied.

V. CONCLUSION

The growing complexity and size of Industrial Internet of Things (IIoT) networks have brought to the fore major limitations around centralized Software-Defined Networking (SDN) models, especially in regards to scalability, responsiveness and energy consumption. This review illustrates that the application of Artificial Intelligence (AI)

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to distributed SDN architectures has a potential future way forward to overcome these shortcomings. In 52 studies reviewed, AI algorithms, including Genetic Algorithms (GA), Reinforcement Learning (RL), Deep Q-Networks (DQN) and swarm-based optimization, all reported measurable performance improvements, of up to 78 % energy savings and 63 percent longer network lifetime, and improvements in adaptability and load balancing.

Nevertheless, there are a few deficiencies of the current studies. The majority of frameworks are only simulated, restricting their usefulness to actual IIoT implementations that must take into account node failure, hardware limitations and harsh operating conditions. Most studies focus on the optimization of a single objective, energy consumption at the expense of multi-objective trade-off including latency, packet delivery ratio, fairness and fault tolerance. In addition, the computational burden of AI algorithms in edge devices is little understood, which casts uncertainty on the possibility of running complex AI models on low-power IIoT devices.

The major future challenges will be controller synchronization in distributed SDN, reduction of reconfiguration overheads in cluster-based routing, and the development of adaptive mobile sink strategies with considerations of interference, deadline-sensitive traffic, and dynamic workloads. Security, interoperability and benchmarking standards are also yet to be explored properly, which restricts the reproducibility and large-scale adoption.

Meanwhile, there are serious opportunities. The lightweight AI models, edge-cloud cooperative systems, and blockchain-based SDN systems will be able to offer robust, decentralized, and secured systems. It is worth developing multi-objective optimization schemes that simultaneously take into account energy consumption, latency, and reliability and are more consistent with the actual needs of IIoT. Furthermore, the establishment of open, trace-based benchmarks and testbeds will be able to speed up reproducibility and practical validation. Future work should also consider explainable AI in SDN controller to enhance trust, transparency, and system debugging in industrial environments.

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